Economically Robust Bitcoin Price Prediction with Machine Learning

A Walk-Forward Study on BTCUSDT 15-Minute Data

Project Report in Scientific Paper Format

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Research Question

Can time-series-validated and economically robust trading signals be derived from BTCUSDT 15-minute data after realistic transaction costs?

Summary

This study investigates whether historical BTCUSDT spot-market data at a 15-minute frequency can be used to derive robust directional forecasts and economically viable trading signals. A complete machine learning pipeline was developed, covering data acquisition, data quality control, exploratory analysis, feature and target engineering, time-series-aware walk-forward validation, baseline comparisons, hyperparameter tuning, probability calibration, and cost-aware threshold and strategy-feasibility analysis. Three horizons were examined: h4, h16, and h96, corresponding to one, four, and 24 hours on 15-minute candles. LightGBM and CatBoost were used as the principal models. Model selection was based on log loss and fold stability; the holdout was used solely for one-time confirmation, while the final test split remained closed throughout the development phase. No Trade was treated as a full alternative to taking a position in the economic evaluation. Limited robust regions existed at low costs for all three horizons, but no candidate overcame the assumed round-trip cost burden of 31.9744 basis points. h16 showed the highest exploratory cost robustness at 10 basis points; h4 and h96 each reached 5 basis points. No horizon passed the Economic Gate. Consequently, no model was approved for final test evaluation or production trading. The central finding is that statistical predictive quality and economic viability must be assessed separately and that a methodologically justified No-Trade result constitutes a valid scientific finding.

Keywords: Bitcoin, BTCUSDT, financial time series, gradient boosting, walk-forward validation, probability calibration, transaction costs, backtest overfitting, No Trade

Abstract

This study investigates whether historical BTCUSDT spot-market data at a 15-minute frequency can be used to derive robust directional forecasts and economically viable trading signals. A complete machine learning pipeline was implemented, covering data acquisition, quality control, exploratory analysis, feature and target engineering, walk-forward validation, baseline models, hyperparameter optimization, probability calibration, and cost-aware threshold and strategy-feasibility analysis. Three prediction horizons were examined: h4, h16, and h96, corresponding to one, four, and 24 hours. LightGBM and CatBoost served as the principal models. Model selection was based on log loss and fold stability; a holdout set was used only for one-time confirmation, while the final test set remained unopened throughout development. No Trade was treated as an explicit alternative in the economic evaluation. Although limited robust regions were identified under low transaction costs, none of the candidates covered the assumed round-trip cost of 31.9744 basis points. The h16 horizon achieved the highest exploratory cost robustness at 10 basis points, whereas h4 and h96 reached 5 basis points. No horizon passed the predefined economic gate, and therefore no model was released for final test evaluation or production trading. The results demonstrate that statistical predictive performance and economic viability must be assessed separately and that a methodologically justified No-Trade outcome constitutes a valid scientific result.

AI Usage Disclosure

AI Usage Disclosure: OpenAI GPT-5.6 Thinking was used to support language editing, document structuring, code review, and visualization drafting. All methodology, implementation decisions, experiments, results, and conclusions were reviewed and validated by the author.

1. Introduction

1.1 Motivation

Cryptocurrency markets combine high liquidity, continuous trading, and pronounced volatility. These properties make Bitcoin an attractive subject for data-driven forecasting methods. At the same time, financial time series are characterized by a low signal-to-noise ratio, non-stationary market regimes, and substantial data-snooping risks. A model may therefore show apparently good classification performance without providing an economic advantage after realistic execution costs. Previous studies report both positive results for machine-learning methods on cryptocurrency data [5, 6] and a strong dependence of profitability on transaction costs and study periods [7].

This study addresses this gap by strictly separating the assessment of statistical predictive quality from economic tradability. Instead of subsequently selecting an attractive backtest configuration, a multi-stage approval procedure is used: models are first selected and calibrated in a time-series-aware manner. Thresholds are then determined within the walk-forward structure and evaluated on outer folds against an explicit No-Trade alternative. Only a candidate that satisfies all statistical and economic conditions would be permitted to open the final test split once.

1.2 Research Question and Contributions

The research question is: Can time-series-validated and economically robust trading signals be derived from BTCUSDT 15-minute data after realistic transaction costs? To answer this question, three prediction horizons are examined and compared within a unified pipeline.

- Development of a modular and reproducible end-to-end pipeline for BTCUSDT 15-minute data.
- Construction of an extensive feature set with 335 numerical variables and explicit leakage protection.
- Use of five chronologically ordered walk-forward folds and a separately locked final test split.
- Comparison of LightGBM and CatBoost, including Optuna tuning and probability calibration.
- Introduction of a cost-aware Economic Gate with No Trade as a genuine decision alternative.
- Documentation of a negative but methodologically robust result without subsequently opening the test set.

Designation	Candles	Time Span	Target
h4	4	1 hour	Direction and forward return after 4 candles
h16	16	4 hours	Direction and forward return after 16 candles
h96	96	24 hours	Direction and forward return after 96 candles

Table 1: Examined prediction horizons.

2. Related Work

2.1 Machine Learning for Cryptocurrency Forecasting

Research on Bitcoin and cryptocurrency forecasting covers classical time-series models, tree-based methods, and neural networks. McNally et al. [6] compared ARIMA with recurrent neural networks for Bitcoin price forecasting and reported advantages of nonlinear methods. Alessandretti et al. [5] examined a large number of cryptocurrencies and showed that simple machine-learning strategies could outperform standard benchmarks under the market conditions prevailing at the time. However, such findings are only partially transferable to high-frequency BTCUSDT data because frequency, liquidity, market regime, and cost model differ substantially.

For financial markets in general, Fischer and Krauss [7] show that complex sequence models can achieve out-of-sample advantages, although profitability may decline substantially after transaction costs and in later time periods. This observation supports the central premise of the present study that statistical forecasting performance must not be equated with economic usability.

2.2 Gradient Boosting, Tuning, and Calibration

LightGBM uses Gradient-based One-Side Sampling and Exclusive Feature Bundling to train gradient boosting efficiently on large and high-dimensional datasets [1]. CatBoost addresses prediction shift, among other issues, through Ordered Boosting and is established as a robust gradient-boosting method [2]. Both models are suitable for the present dataset because they can represent nonlinear interactions and thresholds in heterogeneous technical features.

The hyperparameters were optimized with Optuna, whose define-by-run approach and pruning mechanisms enable efficient search [3]. Because the subsequent trading decision is based on model probabilities, calibration was also examined. Niculescu-Mizil and Caruana [4] show that boosting models in particular may exhibit systematic distortions in their probabilities and that post-hoc calibration methods can improve probabilistic quality.

2.3 Validation and Backtest Overfitting

Time-series data violate the independence assumption of random cross-validation. Bergmeir et al. [8] discuss conditions under which cross-validation can be valid for autoregressive time-series prediction; however, explicitly chronological evaluation remains particularly important for financial forecasting tasks. Bailey et al. [9] show that repeated searches over strategies and parameters can lead to backtest overfitting. The present study reduces this risk through walk-forward folds, a stability penalty, a separate holdout, a closed final test, and a No-Trade benchmark.

3. Data and Data Preparation

3.1 Data Source and Temporal Resolution

The data pipeline processes BTCUSDT spot-market data at a 15-minute interval. The project configuration covers the period from 2020 to 2026; historical archives are consolidated with current API data. Each candle contains at least open, high, low, close, volume, timestamps, and additional trade-related information. All timestamps are normalized to UTC and stored in a canonical Parquet dataset.

3.2 Data Quality Gate

A data quality gate is executed before every analysis. It checks schema and data types, missing values, duplicates, chronological order, candle intervals, OHLC consistency, negative values, zero volume, trade counts, and timestamp logic. Critical errors block downstream modules; tolerable anomalies are documented as warnings. This prevents inconsistent candles or silent data gaps from entering feature calculation and model training.

3.3 Feature Engineering

The dataset used for model validation comprises 335 numerical features. The variables represent short- and medium-term market dynamics and are calculated exclusively from information known at the prediction time. Target-like variables are explicitly excluded from the feature set.

Feature Group	Examples	Purpose
Returns and Momentum	simple/logarithmic returns, absolute and relative momentum measures	directional and trend information
Volatility and Risk	rolling standard deviation, range, downside measures	market regime and uncertainty

Feature Group	Examples	Purpose
Technical Indicators	RSI, MACD, moving averages, Bollinger-oriented measures	nonlinear price states
Volume and Trades	volume changes, quote volume, trade count	activity and liquidity
Candle Structure	body, wicks, high-low range, relative position	candle microstructure
Time Features	hour, weekday, cyclical encodings	seasonal trading patterns
Lags and Rolling Features	lagged indicators and multi-window aggregates	temporal dependencies

Table 2: Main groups of the feature set.

3.4 Target Variables

For each horizon, both a binary directional target and a continuous forward return are generated. The binary target is used by the probabilistic classification model; the continuous return enables the subsequent net-return evaluation of thresholds.

$$r(t,h) = P(t+h) / P(t) - 1$$
$$y(t,h) = 1 \text{ if } r(t,h) > 0; \text{ otherwise } 0.$$

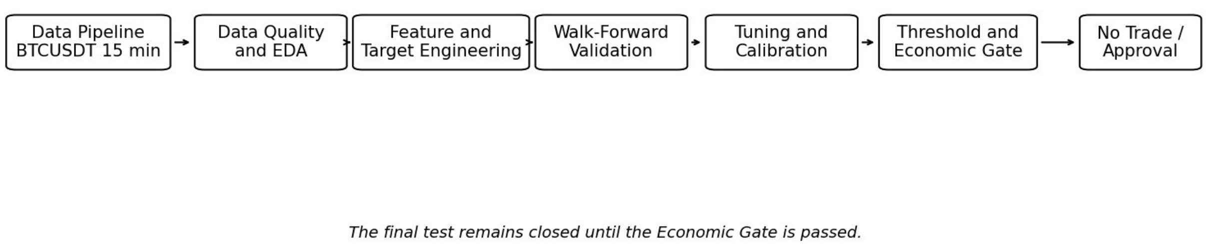

Figure 1: Methodological pipeline and approval logic.

4. Methodology

4.1 Time-Series-Aware Data Splitting

The evaluation uses five walk-forward folds with an expanding training window. Each fold trains on historical observations and evaluates on a later temporal segment. Horizon-dependent separation gaps prevent overlapping forward targets from carrying information across fold boundaries. In addition, a holdout area with separate training and validation portions and a final test split are defined. In the validated h16 snapshot, the holdout training area comprised 137,976 rows and the holdout validation area 35,104 rows. The test split existed but was not opened.

Component	Use	Selection Permitted?
Walk-Forward Training	model fitting and inner threshold selection	Yes
Walk-Forward Validation	model comparison, calibration, and outer economic evaluation	Yes, according to predefined rules
Holdout Validation	one-time confirmation of an already selected tuning candidate	No, no subsequent ranking change
Final Test	one-time final evaluation of a fully approved candidate	Only after passing the Economic Gate

Table 3: Roles of the temporal data segments.

4.2 Modeling and Hyperparameter Tuning

LightGBM and CatBoost are trained as probabilistic binary classifiers. The primary optimization metric is log loss because it evaluates not only class assignment but also the quality of the output probabilities. To penalize unstable solutions, a proportion of the standard deviation is added to the mean fold metric. The search spaces include tree depth, learning rate, number of trees, regularization, subsampling, and feature subsampling, among other parameters.

$$LogLoss = -(1/N) * Sum[y_i \ln(p_i) + (1-y_i) \ln(1-p_i)]$$
$$Tuning\ Score = mean\ log\ loss + 0.25 * fold\ standard\ deviation.$$

4.3 Probability Calibration

After tuning, raw probabilities and sigmoid calibration are compared. The calibration method is selected exclusively on the basis of the walk-forward folds. A holdout must not be used to select the calibration method. The final model-calibration combination may therefore differ from the pure tuning ranking if another model produces more consistent or better-calibrated probabilities.

4.4 Nested Threshold Selection

The trading threshold is not optimized globally on all validation data. Instead, selection is nested within the temporal folds. For each candidate, the resulting trades are evaluated after costs using the continuous forward returns. No Trade receives a minimum objective value of zero. A threshold is executable only if it outperforms No Trade and satisfies all minimum conditions. Otherwise, only a diagnostic reference threshold is stored, which may neither be traded nor applied to the final test.

4.5 Cost Model and Economic Gate

The cost model accounts for transaction fees, spread, and slippage. The round-trip cost burden stored in the final artifacts is 31.9744 basis points. In addition, cost levels of 0, 5, 10, 20, 31.9744, and 40 basis points are examined to determine a signal's robustness limit. The Economic Gate requires not only a positive mean return but also several stability conditions to be satisfied simultaneously.

Gate Criterion	Meaning
Proportion of Tradable Folds	The signal must not occur only in isolated time segments.
Proportion of Positive Outer Folds	Positive results must be broadly distributed over time.
Minimum Number of Trades	Very small samples are not accepted as robust.
Positive Net Return	Returns are evaluated after the complete cost model.
Net-Return LCB > 0	The conservative lower confidence bound must be positive.

Gate Criterion	Meaning
Profit Factor > 1	Profits must exceed losses.
Positive Stability Score	The mean and fold dispersion are considered jointly.

Table 4: Central conditions of the Economic Gate.

5. Experimental Setup

5.1 Software and Reproducibility

The final execution was performed on Linux with Python 3.14.5 in a virtual environment. The pipeline stores configurations, parameter sources, models, fold metrics, calibration diagnostics, threshold decisions, and feasibility reports as versionable artifacts. The final aggregator performs no new learning or optimization steps; instead, it generates comparison tables, a final JSON file, a Markdown conclusion, and a manifest with SHA-256 checksums.

5.2 Frozen Reference Runs

Horizon	Reference Run	Special Characteristic
h4	reports/threshold_signal_optimization/20260729T202947619086Z	Older generic report structure
h16	reports/h16_screening/h16_20260730T022812	Complete screening orchestrator
h96	reports/h96_screening/h96_20260730T075624	Complete screening orchestrator

Table 5: Project runs frozen for the final report.

5.3 Evaluation Metrics

The statistical evaluation focuses on log loss, the fold mean, and the fold standard deviation. Holdout metrics are additionally considered for one-time confirmation. The economic evaluation uses trade count, net returns, profit factor, fold proportions, and a conservative lower confidence bound. This separation prevents a small improvement in a classification metric from being interpreted automatically as a trading advantage.

6. Results

6.1 Statistical Model and Calibration Results

For h16, tuning selected CatBoost with a mean walk-forward log loss of 0.672113 and a standard deviation of 0.006467. The subsequent joint calibration evaluation confirmed CatBoost with sigmoid calibration and a mean log loss of 0.663146 with a standard deviation of 0.018843. h16 therefore showed the clearest separation from the 50/50 reference of approximately 0.6931.

For h96, the tuned models were close to the random reference. CatBoost achieved a mean log loss of 0.693591 with a standard deviation of 0.001188 during tuning; LightGBM was nearly tied at 0.693615. However, the calibration evaluation selected LightGBM with raw probabilities. The mean log loss was 0.692885 with a very small fold dispersion of 0.000371. The statistical advantage therefore remained small.

The h4 run originates from an older report structure. The final artifacts identify LightGBM with sigmoid calibration as the examined combination. Because the intermediate reports were not fully transferred into the common tabular structure used later, no retrospectively harmonized CV values are reported for h4. This limitation does not affect the final threshold and cost decision.

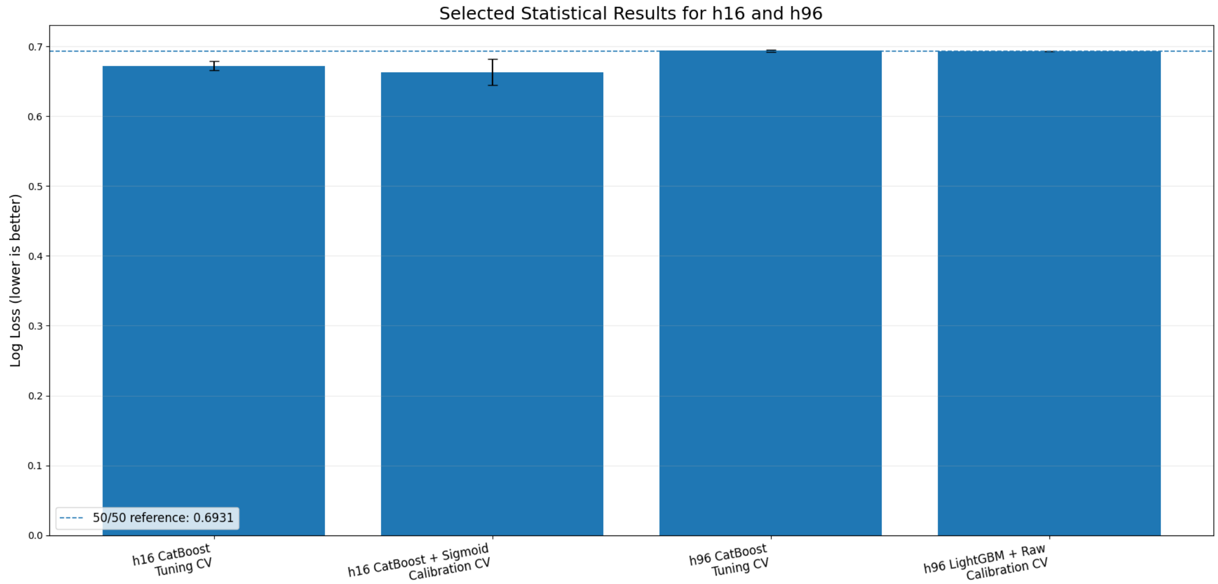

Figure 2: Selected log-loss results for h16 and h96; error bars show the fold standard deviation.

Horizon	Tuning Selection	Calibration Selection	CV Log Loss	Interpretation
h4	LightGBM	LightGBM + Sigmoid	Legacy structure; not harmonized	Final economic decision available
h16	CatBoost	CatBoost + Sigmoid	0.663146 +/- 0.018843	Strongest statistical candidate
h96	CatBoost narrowly in tuning	LightGBM + Raw	0.692885 +/- 0.000371	Only marginally better than 50/50

Table 6: Final model and calibration selection.

6.2 Threshold and Economic-Gate Results

No executable threshold was approved for any horizon. For h4, all five folds selected No Trade; the diagnostic reference threshold was 0.485. In h16, all five folds were likewise classified as No Trade. The reference threshold of 0.428497 was purely diagnostic; trade-fold proportion, positive outer-fold ratio, trade count, net return, profit factor, and stability score failed to meet the minimum conditions. h96 also resulted in No Trade in every fold.

This result is stricter than a simple backtest evaluation. A model can produce probabilities that are slightly informative statistically without a threshold reliably generating positive trades after costs. The Economic Gate prevents such a weak candidate from being approved because of isolated positive periods or a subsequently selected threshold.

6.3 Cost Robustness

The feasibility analysis identified robust regions at lower costs for all horizons. h4 and h96 remained robust up to a maximum of 5 basis points, while h16 remained robust up to 10 basis points. The assumed current round-trip cost burden of 31.9744 basis points was not covered by any candidate. h16 was therefore relatively strongest but covered only approximately 31 percent of the assumed costs.

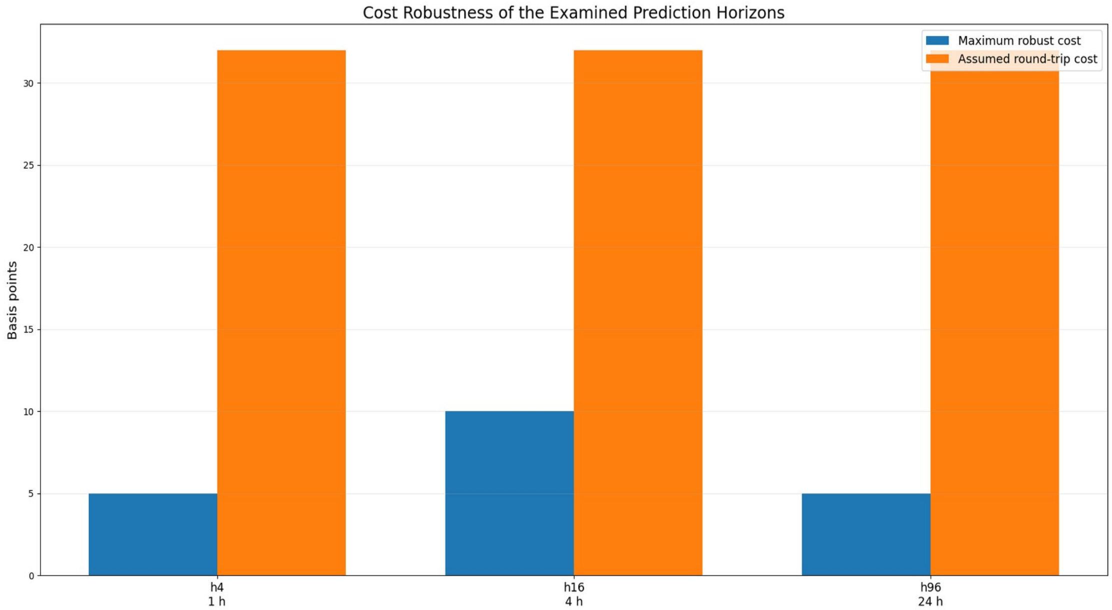

Figure 3: Maximum robust costs compared with the assumed round-trip cost burden.

Horizon	Final Model	Calibration	Decision	Max. Robust Cost	Current Costs Viable
h4	LightGBM	Sigmoid	No Trade	5 bp	No
h16	CatBoost	Sigmoid	No Trade	10 bp	No
h96	LightGBM	Raw	No Trade	5 bp	No

Table 7: Final economic results.

6.4 Status of the Final Test Set

The final test split was not evaluated for any horizon. The tuning, calibration, threshold, and screening artifacts explicitly state that the test was used neither for selection nor for tuning. Because no candidate passed the Economic Gate, opening the test was methodologically impermissible. The absence of a final test metric is therefore not a missing project step but part of the predefined research design.

7. Discussion

7.1 Statistical Signal versus Economic Viability

The results illustrate the difference between probabilistic model quality and an implementable trading strategy. h16 showed a log loss clearly below the 50/50 reference, yet economic threshold selection produced no approved trade candidate in any outer fold. This can occur when probability shifts are systematic but too small, when directional advantages are not accompanied by sufficient forward returns, or when profitable periods are temporally unstable.

For h96, statistical discrimination was already weak. The selection of LightGBM with raw probabilities shows that an additional sigmoid transformation offered no consistent advantage. The result near the random reference is consistent with the economic No-Trade decision. h4 showed a limited robust region at very low costs but likewise came nowhere near covering the realistic cost assumption.

7.2 Importance of the Horizon

The four-hour horizon h16 proved to be the strongest compromise. Compared with h4, a longer horizon reduces the relative dominance of short-term microstructure and execution costs. Compared with h96, the target remains more closely related to the technical and momentum features, whereas the 24-hour horizon may be influenced more strongly by unobserved news, macroeconomic factors, and regime changes. Nevertheless, h16 was also not economically robust enough.

7.3 No Trade as a Scientific Result

Many forecasting projects treat a position as a mandatory model output. As a result, a trade is generated even when probabilities are uncertain. In contrast, the present pipeline uses No Trade as the economic null alternative. This decision raises the requirements but prevents artificial profitability through forced transactions. A completely negative gate result is scientifically informative: under the examined conditions, no sufficiently robust edge exists for approval.

7.4 Relationship to the Literature

The results do not directly contradict positive findings from earlier cryptocurrency studies. Alessandretti et al. [5] examined daily data from many cryptocurrencies at an earlier stage of the market, whereas the present study uses high-frequency BTCUSDT data and a stricter cost and validation regime. McNally et al. [6] focused primarily on price forecasting and model comparison rather than a multi-stage Economic Gate. By contrast, the decline in advantage after costs observed here is consistent with the sensitivity of financial machine-learning strategies to market periods and transaction costs documented by Fischer and Krauss [7].

8. Limitations

The explanatory scope of the study is tied to the specific data basis and the selected research design. The main limitations are:

- BTCUSDT spot data from a single market microstructure were examined; futures, other exchanges, and other trading pairs may differ.
- The cost model is realistic but static. Actual execution costs depend on order type, size, liquidity, fee tier, and latency.
- The feature set is based predominantly on price, volume, and technical variables. Order-book, on-chain, derivatives, news, and macroeconomic data were not integrated.
- The model class is limited to tabular gradient-boosting methods and baselines; sequence models and regime-switching approaches were not examined conclusively.
- The h4 artifacts originate from an older report structure. The final decision is reproducible, but not all intermediate metrics are directly harmonized with h16 and h96.
- The cost-feasibility analysis is diagnostic. A positive region at low costs would have to be confirmed in a newly specified pipeline before test approval.
- Market regimes may change. Robust historical walk-forward performance does not guarantee future stability.

9. Conclusion and Future Work

9.1 Conclusion

The research question must be answered negatively for the examined pipeline. Although limited statistical signals and robust regions at low costs could be derived from the BTCUSDT 15-minute data, no candidate overcame the assumed round-trip cost burden of 31.9744 basis points. h16 was the strongest horizon both statistically and economically, but its robustness limit of 10 basis points remained clearly below the required level. h4 and h96 each reached 5 basis points.

The final decision for all horizons was No Trade. The Economic Gate was not passed, the final test split remained closed, and no production approval was granted. The study therefore fulfills its methodological objective: it provides not only a forecasting model but also a reproducible decision logic that consistently rejects weak or cost-limited signals.

9.2 Future Work

A new research phase should not continue through further optimization on the outer folds already analyzed. Instead, new data sources and a prespecified confirmatory pipeline are required. The following directions appear particularly relevant:

- Order-book and microstructure features for explicitly modeling spread, liquidity, and short-term order flow.
- Derivatives-market data such as funding rates, open interest, and liquidations.
- On-chain, news, sentiment, and cross-asset variables.
- Regime-dependent models and dynamic selection of the prediction horizon.
- Long/No-Trade targets that already incorporate costs into the target definition.
- Maker/taker scenarios and realistic limit-order execution.
- Paper trading as an additional genuine out-of-sample phase before any use of real capital.

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Appendix A: Reproducibility

The following commands constitute the minimum finalization workflow. They assume that the project structure and the frozen report directories are available.

```
cd "/home/milan/uni/python/BTC TSP/BTC TSP 15min"
source "/home/milan/uni/python/BTC TSP/.venv/bin/activate"
python final_project_report.py validate
python final_project_report.py run --h4-run-dir
"reports/threshold_signal_optimization/20260729T202947619086Z" --h16-run-dir
"reports/h16_screening/h16_20260730T022812" --h96-run-dir
"reports/h96_screening/h96_20260730T075624"
```

```
python final_project_report.py status
```

Appendix B: Authoritative Final Artifacts

Artifact	Function
reports/final/final_summary.json	Machine-readable overall summary
reports/final/final_conclusion.md	Fully formulated scientific conclusion
reports/final/horizon_comparison.csv	Comparison of h4, h16, and h96
reports/final/model_comparison.csv	Model and tuning overview
reports/final/cost_feasibility_comparison.csv	Cost robustness and viability
reports/final/source_runs.json	Frozen reference runs
reports/final/artifact_manifest.json	Paths, sizes, timestamps, and SHA-256 checksums
10_final_project_results.ipynb	Visual summary of the final results

Table 8: Authoritative sources for project finalization.